# Machine Learning

Introduction (continued)

# Resources: Journals

- Journal of Machine Learning Research www.jmlr.org
- Machine Learning
- IEEE Transaction on Neural Networks
- IEEE Transaction on Pattern Analysis and Machine Intelligence
- Annals of Statistics
- Journal of the American Statistical Association

# Resources: Conferences

- International Conference on Machine Learning (ICML)
- European Conference on Machine Learning (ECML)
- Neural Information Processing Systems (NIPS)
- Computational Learning
- International Joint Conference on Artificial Intelligence (IJCAI)
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
- IEEE Int. Conf. on Data Mining (ICDM)

# Framework

- 4 for Deep Learning:
  - TensorFlow (+Keras), MXNet, caffe, Theano
- 5 for generic machine learning:
  - scikit-learn, Spark MLLib, mlpack, Accord.NET Framework, Weka/Meka
- 3 for cloud (actually those are not frameworks):
  - Amazon Machine Learning
  - Google Cloud Machine Learning
  - Azure ML Studio

# Covered Topics

- Concept learning
- Linear Regression
- Logistic Regression
- Naïve Bayes
- Decision Tree
- Support Vector Machine
- K-nearest Neighbors
- Neural Network
- K-means
- Feature Reduction
- ML Techniques
- Text Mining
- Big Data Framework

# Classification Tasks

- Binary classification
- Multi-class classification
  - One-Vs-Rest
- Multi-label classification
  - Binary Relevance

# Concept Learning and The General-To-Specific Ordering

# Concept Learning

- Acquire **general** concepts from **specific** training examples.
- Each such concept can be viewed as describing some subset of objects or events defined over a larger set.

# Concept Learning Task

- Target Concept: Days on which my friend enjoys his water sport (Attribute: EnjoySport)
- Task: Predict the value of EnjoySport based on the values of other attributes.
- Positive and negative training examples for the target concept EnjoySport

| Example | Sky | AirTemp | Humidity | Wind | Water | Forecast | EnjoySport |
|---|---|---|---|---|---|---|---|
| 1 | Sunny | Warm | Normal | Strong | Warm | Same | Yes |
| 2 | Sunny | Warm | High | Strong | Warm | Same | Yes |
| 3 | Rainy | Cold | High | Strong | Warm | Change | No |
| 4 | Sunny | Warm | High | Strong | Cool | Change | Yes |

# Concept Learning Task

- Hypothesis Representation
  - A simple representation: each hypothesis consists of a conjunction of constraints on the instance attributes.
  - Each hypothesis be a vector of six constraints.
    - "**?**" indicate any value is acceptable for this attribute
    - A single required value for the attribute
    - **ϕ** indicate that no value is acceptable
  - Example
    - <?, Cold, High, ?, ?, ?>
    - <?, ?, ?, ?, ?, ?>
    - < ϕ, ϕ, ϕ, ϕ, ϕ, ϕ>

# Notations

- Instance $\boldsymbol{x}$
- A set of instances $\boldsymbol{X}$
- Target Concept $\boldsymbol{c}$: The concept or function to be learned
- Hypothesis $\boldsymbol{h}$: a Boolean-valued function defined over $\boldsymbol{X}$. The goal of the learner is to find a hypothesis $\boldsymbol{h}$ such that $\boldsymbol{h(x)=c(x)}$
- Hypothesis Set $\boldsymbol{H}$
- Positive examples
- Negative examples
- Training dataset $\boldsymbol{D}$

# Concept Learning Task

- Given
  - Instances X: Possible days, each described by the attributes
    - *Sky (with possible values Sunny, Cloudy, and Rainy)*
    - *AirTemp (with values Warm and Cold)*
    - *Humidity (with values Normal and High)*
    - *Wind (with values Strong and Weak)*
    - *Water (with values Warm and Cold), and*
    - *Forecast (with values Same and Change).*
  - Hypotheses ***H***: Each hypothesis is described by a conjunction of constraints on the attributes Sky, AirTemp, Humidity, Wind, Water and Forecast. The constraints may be "?" , ϕ or a specific value.
  - Target Concept ***c***: EnjoySport: X→ {0,1}
  - Training Examples ***D***: Positive and negative examples of the target function
- Determine
  - A hypothesis ***h*** in ***H*** such that ***h(x)=c(x)*** for all ***x*** in ***X***

# Inductive Learning Hypothesis

- Inductive Learning
  - Determine a hypothesis ***h*** over set of instances ***X***
- Inductive
  - Guarantee that the output hypothesis fits the target concept over the training data
- Fundamental Assumption
  - Hypothesis that best fits the observed training data
- The Inductive Learning Hypothesis
  - Any hypothesis found to approximate the target function well over a sufficiently large set of training examples will also approximate the target function well over unobserved examples.

# General-to Specific Ordering of Hypotheses

- Two hypotheses
  - $h_1$=<Sunny, ?, ?, Strong, ?, ?>
  - $h_2$=<Sunny, ?, ?, ?, ?, ?>
  - Any instance classified positive by $\boldsymbol{h_1}$ will also be classified positive by $\boldsymbol{h_2}$, thus $\boldsymbol{h_2}$ is more general than $\boldsymbol{h_1}$

# Find-S Algorithm

- Initialize ***h*** to the most specific hypothesis in ***H***
- For each positive training instance ***x***
  - For each attribute constraint $a_i$ in h
    - If the constraint $a_i$ is satisfied by ***x***
    - Then do nothing
    - Else replace $a_i$ in ***h*** by the next more general constraint that is satisfied by ***x***
- Output hypothesis ***h***

# Find-S: Finding a Maximally Specific Hypothesis

- Find-S on EnjoySport Task
  - $\boldsymbol{h_0} \leftarrow$<ϕ, ϕ, ϕ, ϕ, ϕ, ϕ>
  - $\boldsymbol{h_1} \leftarrow$<Sunny, Warm, Normal, Strong, Warm, Same>
  - $\boldsymbol{h_2} \leftarrow$<Sunny, Warm, ?, Strong, Warm, Same>
  - $\boldsymbol{h_3} \leftarrow$ The algorithm makes no change to $\boldsymbol{h}$ when encountering a negative example
  - $\boldsymbol{h_4} \leftarrow$<Sunny, Warm, ?, Strong, ?, ?>

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# Find-S: Finding a Maximally Specific Hypothesis

- Find-S is guaranteed to output the most specific hypothesis within ***H*** that is consistent with the positive training examples.
  - Why anyways prefer the most specific hypothesis? Maybe a more general or something in between could be better.
  - The algorithm finds and returns only one hypothesis. But there usually are other hypotheses consistent with the training data. Why discard them?

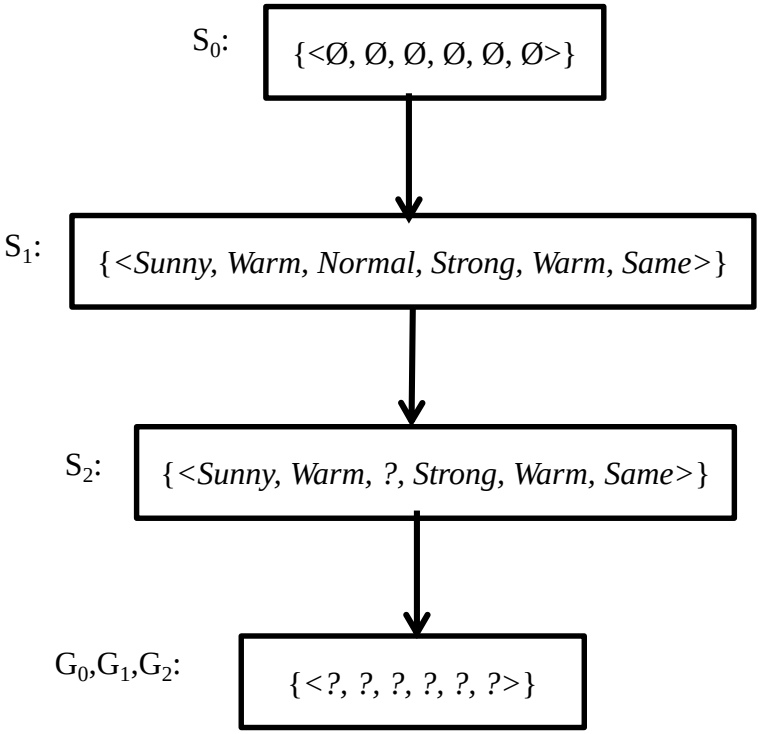
# Candidate-Elimination Algorithm

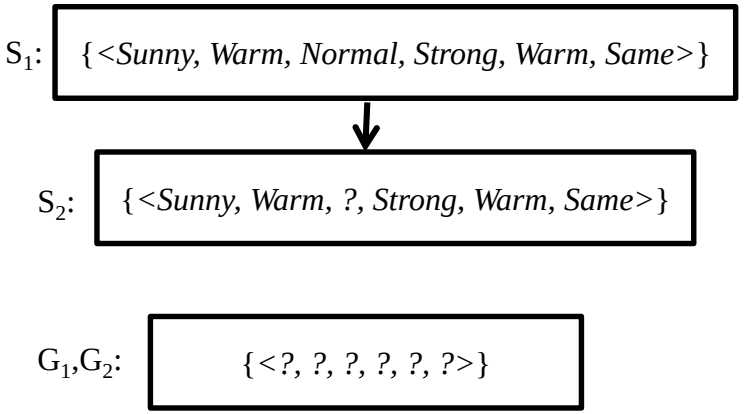
- Find-S only output one of the correct hypotheses
- The output of Candidate-Elimination Algorithm are all the hypotheses consistent with training data.
- Initialize ***G*** to the set of maximally general hypotheses in ***H***
- Initialize ***S*** to the set of maximally specific hypotheses in ***H***

# Candidate-Elimination Algorithm

- For each training example d, do
  - If ***d*** is a ***positive*** example
    - Remove from ***G*** any hypothesis inconsistent with ***d***
    - For each hypothesis ***s*** in ***S*** that is not consistent with ***d***
      - Remove ***s*** from ***S***
      - Add to ***S*** all minimal generalizations ***h*** of ***s*** such that
        - » ***h*** is consistent with ***d***, and some member of ***G*** is more general than h
      - Remove from ***S*** any hypothesis that is more general than another hypothesis in ***S***
  - If ***d*** is ***negative*** example
    - Remove from ***S*** any hypothesis inconsistent with ***d***
    - For each hypothesis ***g*** in ***G*** that is not consistent with ***d***
      - Remove ***g*** from ***G***
      - Add to ***G*** all minimal specializations ***h*** of g such that
        - » ***h*** is consistent with ***d***, and some member of ***S*** is more specific than ***h***
      - Remove from ***G*** any hypothesis that is less general than another hypothesis in ***G***

# An Illustrative Example




Training examples:

1. *<Sunny, Warm, Normal, Strong, Warm, Same>, Enjoy Sport=Yes*
2. *<Sunny, Warm, High, Strong, Warm, Same>, Enjoy Sport=Yes*

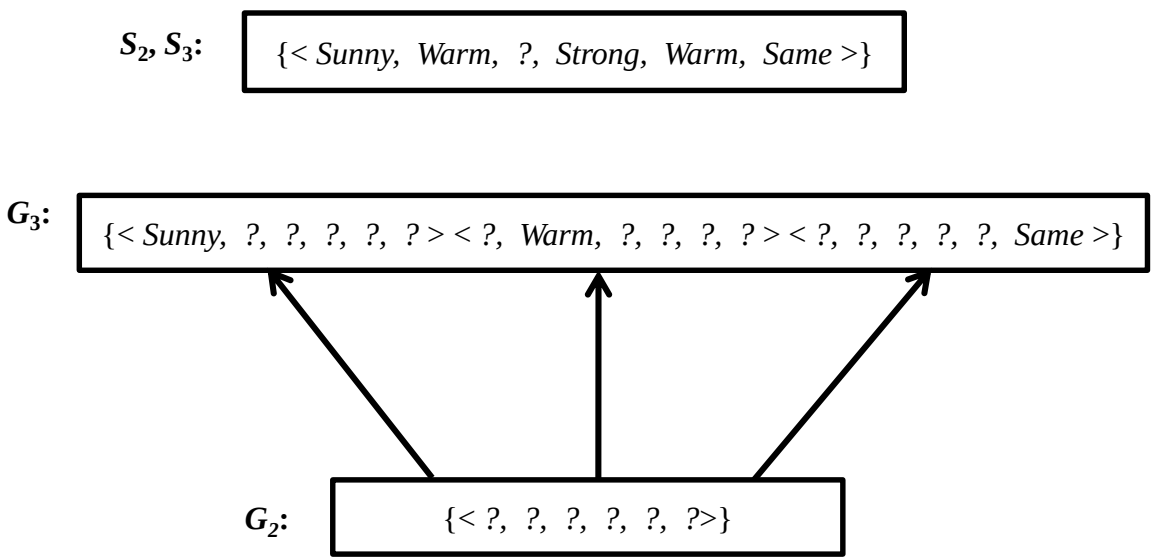
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$S_3$ : {< *Sunny, Warm, ?, Strong, Warm, Same* >}

$S_4$ : {< *Sunny, Warm, ?, Strong, ?, ?* >}

$G_4$ : {< *Sunny, ?, ?, ?, ?, ?* > < *?, Warm, ?, ?, ?, ?* > }

$G_3$ : {< *Sunny, ?, ?, ?, ?, ?* > < *?, Warm, ?, ?, ?, ?* > < *?, ?, ?, ?, ?, Same* >}

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